

COVID-19 Detection from Chest X-Rays

Project report

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Abstract

The work presented in this report aims to develop a neural network capable of distinguishing COVID-19 pneumonia from other causes that may lead to pneumonia. To this end, the chest X-ray images used as the dataset were analyzed using a transfer learning approach. In particular, the DenseNet121 architecture, pre-trained on the ImageNet dataset, was used, with its final layer replaced to perform the desired classification task.

Initially, a three-class classification was investigated, distinguishing between COVID-19 pneumonia, bacterial pneumonia, and viral pneumonia. Due to the limited number of cases of viral pneumonia, a binary classification was subsequently adopted, using COVID and NO COVID as the two classes.

In addition to evaluating performance metrics such as the F1-score, ROC-AUC curve, confusion matrix, and accuracy, the analysis also aimed to investigate which regions of the input image had the greatest influence on the network's classification. This analysis was performed by studying the corresponding saliency maps.

1. Dataset

The chest radiographs used in this work were obtained from the COVID-19 Image Data Collection¹. The dataset contains 930 chest X-ray images of individuals affected by pneumonia.

1.2. Three-Class Classification

From the initial dataset, 697 images were selected based on their point of view as anteroposterior (AP), posteroanterior (PA), or anteroposterior supine (AP Supine), as these are the only image categories that can be processed by the selected lung segmentation algorithm (see Section 2).

The final dataset, consisting of 192 images classified as pneumonia not caused by COVID-19 and 505 images of COVID-19 pneumonia, was divided into a training set and a test set. A total of 402 images from patients with COVID-19 were used for training, while 179 images corresponding to cases not caused by COVID-19 were included in the training set; the remaining images were used to construct the test set. Furthermore, the data were split in such a way that all chest X-ray images belonging to the same patient were assigned to the same subset.

¹ COVID-19 image data collection, arXiv:2003.11597, 2020
Joseph Paul Cohen and Paul Morrison and Lan Dao
<https://github.com/ieee8023/covid-chestxray-dataset>

1.3. Binary Classification

The total number of selected images was 670. This selection resulted from an additional preprocessing step to identify images suitable for subsequent processing (see Section 5.4). The selected images were subsequently divided into a training set of 567 images and a test set of 103 images. The training set comprised 424 images of COVID-19 pneumonia and 143 images of other types of pneumonia, whereas the test set included 77 images of COVID-19 pneumonia and 26 images classified as non-COVID-19.

2. Lung Segmentation Algorithm

Once the images had been selected, a lung mask was generated for each chest X-ray image. This is a common practice in this type of work, as it allows the network to focus exclusively on the lung region.

The segmentation algorithm used in this work is the one described in the following paper: *Lung Segmentation from Chest X-rays using Variational Data Imputation*, by Raghavendra Selvan et al., 2020².

The algorithm combines a U-Net architecture with a Variational AutoEncoder (VAE). The VAE learns, from normal chest X-ray images, a latent representation of the image features, which can be used to handle missing or corrupted information. The U-Net then exploits this representation to perform lung segmentation. During the testing phase, highly opacified chest X-ray images are treated as incomplete data, allowing the model to infer the missing information and obtain the lung segmentation even under more challenging conditions.

The output of the algorithm is a binary mask corresponding to the lung region, in which the lungs are represented in white and the entire area outside the lungs is represented in black.

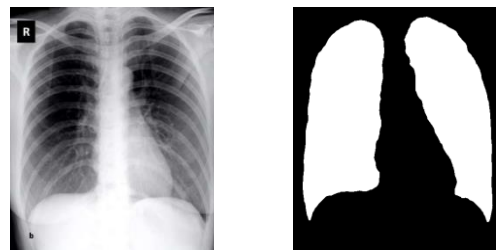


Figure 1: Chest X-ray image and corresponding lung segmentation of a patient with COVID-19

² <https://arxiv.org/abs/2005.10052>

3. Model training

The same training approach was adopted for both classification tasks, based on the use of DenseNet121, a deep Convolutional Neural Network (CNN) pre-trained on the ImageNet dataset. Before being fed into the network, the images and their corresponding masks were converted to grayscale and resized to (224 x 224). Since DenseNet121 requires three-channel input images, the single grayscale channel was replicated three times, resulting in a three-channel image. The masks were then converted into binary masks, while the images were normalized to the [0,1] range.

In addition, a background randomization procedure was applied with the aim of reducing the network's dependence on background characteristics.

For both classification tasks, the original final classifier of DenseNet121 was replaced with a linear layer, whose number of output neurons was defined according to the number of classes considered. The classifier receives a 1024-dimensional feature vector produced by the convolutional part of DenseNet121 and returns the logits associated with the different classes. These values are unnormalized scores associated with each class.

Cross-Entropy Loss was used as the loss function during training. The loss function internally applies the log-softmax operation to the logits. The network weights were updated using the Adam optimizer, with a learning rate of 10^{-4} . The model was trained using a batch size of 8 images for a maximum of 50 epochs.

To reduce the risk of overfitting, an early stopping mechanism with a patience of 10 epochs was introduced. Training was therefore stopped when the validation loss did not improve for 10 consecutive epochs. During training, the model corresponding to the lowest validation loss achieved was also saved. At the end of training, the weights corresponding to this model were restored.

The outputs of the training procedure are the trained model, the validation accuracy over the training epochs, and a plot showing the evolution of the validation loss and training loss as a function of the number of epochs.

In addition, data augmentation was applied exclusively to the training set, with the aim of increasing the amount of data available during training. The transformations included random rotations between -8° and $+8^\circ$, translation between -5 and $+5$ pixel and brightness and contrast variations ranging from 0.9 to 1.1. These transformations introduced variability into the images while preserving their main characteristics.

3.2. Three-Class Classification

For the three-class classification task, the original *metadata.csv* file, containing 17 labels, was used, excluding samples that could not be assigned to either the BACTERIA or VIRAL categories. The different labels were grouped into three categories: BACTERIA, COVID, and VIRAL. Images associated with diagnoses containing "COVID-19", "SARS", or "MERS-CoV" were assigned to the COVID class. The BACTERIA class included the labels "Streptococcus", "Klebsiella", "Escherichia coli", "Nocardia", "Mycoplasma", "Legionella", "Chlamydomphila", "Staphylococcus", and "bacterial". In the end, the VIRAL class included the labels "virus", "viral", "Influenza", "Varicella", and "Herpes".

Of the 697 selected images, 580 were divided into training and validation sets according to a 4:1 ratio. Since the COVID class contained more images than the BACTERIA and VIRAL classes, data augmentation was applied exclusively to the training images belonging to these two classes, with the aim of balancing the classes. Specifically, the BACTERIA and VIRAL classes were increased to match the number of images in the COVID class. In this configuration, the final classifier of the DenseNet121 consists of three output neurons, corresponding to the BACTERIA, COVID, and VIRAL classes. The network has a total of 6,956,931 parameters, all of which are trainable during the training process.

3.3. Binary Classification

For the binary classification task, the training scheme is described in Section 3. The categories in the original *metadata.csv* file were grouped into two classes: COVID and NO COVID. Images associated with *finding* column containing "COVID-19", "SARS", or "MERS-CoV" were classified as COVID, whereas all other categories were assigned to the NO COVID class.

Of the 670 selected images, 567 were divided into training and validation sets according to a 4:1 ratio. Since the NO COVID class contained fewer images than the COVID class, the data augmentation procedure was applied exclusively to NO COVID images belonging to the training set.

During the splitting procedure, the *patientid* column was used as a grouping criterion to ensure that images belonging to the same patient were not simultaneously included in the two subsets, to reduce the risk of *data leakage*.

For the binary classification task, the final DenseNet121 classifier was configured with two output neurons, one for each class. The network contains a total of 6955906 parameters, all of which were trainable during the training process.

4. Model Performance Evaluation

The test code was used both to evaluate the performance of the classifier and to analyze the image regions that are most relevant to the classification. The images were preprocessed according to the procedure described for the training phase and these images were then fed into the trained model. Classifier performance was evaluated by comparing the obtained predictions with the true labels using accuracy (A), precision (P), recall (R), F1-score (F1), the confusion matrix and the ROC curve with the corresponding ROC-AUC.

To analyze the model's behavior and identify the image regions most involved in the classification, a saliency map analysis was also performed. For each image, separate maps were computed for the different classes in order to highlight the regions to which the network output is most sensitive. The saliency maps were obtained from the gradient of the logits with respect to the image pixels.

Based on the lung mask, the image was subsequently divided into four regions: the *lung core*, corresponding to the inner part of the lungs; the *inner border*, consisting of the inner part adjacent to the lung boundary; the *outer border*, corresponding to the part immediately outside the lungs; and the *background*, consisting of the remaining external area. For each region, the percentage of saliency relative to the total saliency of the image was calculated in order to quantify the relative contribution of the different areas to the model's output. For the binary classification task, the overall *outside lung* saliency component was also calculated by summing the contributions of the outer border and background regions.

In the end, to evaluate the possible influence of the background on the model's performance and behavior, different image-processing configurations were considered. In particular, two configurations were compared: a black background, obtained by performing an element-wise multiplication between the original image and the binary mask, and a randomized background, obtained by assigning random values between 0 and 255 to the external region. The latter configuration was introduced to reduce the possibility that the network would learn background specific features and to assess whether the classification primarily relied on information contained within the lungs or on the immediately surrounding regions.

5. Results

This chapter presents the main results obtained during both the training and validation phases for the two classification tasks.

5.2. Three-Class Classification: Training Results

The following graphs show the results obtained from the final version of the three-class classifier:

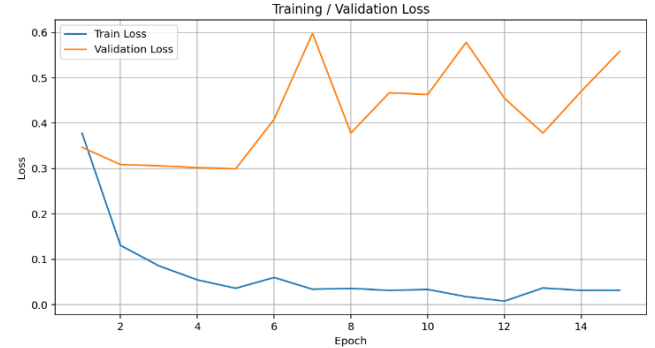


Figure 2: Training e validation loss of the three-class classifier obtained using the model with a randomized background

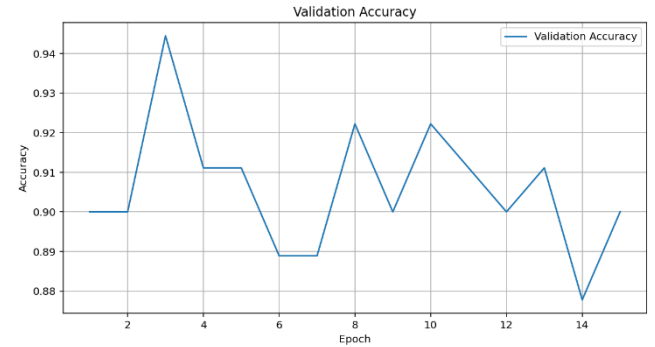


Figure 3: Validation accuracy of the three-class classifier obtained using the model with a randomized background

As can be observed, the model's results are not optimal. In contrast to the regular decrease in training loss, the validation loss exhibits a fluctuating trend, with a relatively large gap between the two. This is indicative of an overfitting condition, likely due to the insufficient number of samples in the dataset not associated with the COVID-19 class. A validation accuracy of 0.9 was also recorded. The best-performing model was the one obtained at the fifth epoch.

5.3. Three-Class Classification: Validation Results

The initial test results, obtained using a black background, showed performance above that of a random classifier. However, the performance metrics revealed some irregularities (see Tables 1 and 2). The ROC curve (see Fig. 4) exhibits an anomalous trend for the VIRAL class, while the corresponding P, R, and F1 values are zero. This is likely due to the presence of only a single sample belonging to this class in the test set. Furthermore, the R and F1 scores for the BACTERIA class are extremely low compared to those of the COVID class.

	Precision	Recall	F1 score	support
BACTERIA	1.00	0.08	0.15	12
COVID	0.90	1.00	0.94	103
VIRAL	0.00	0.00	0.00	1

Table 1: The three-class classification model's performance metrics, by using black background

		PREDICT		
		BACTERIA	COVID	VIRAL
REAL	BACTERIA	1	11	0
	COVID	0	103	0
	VIRAL	0	1	0

Table 2: The three-class classification model's confusion matrix, by using black background

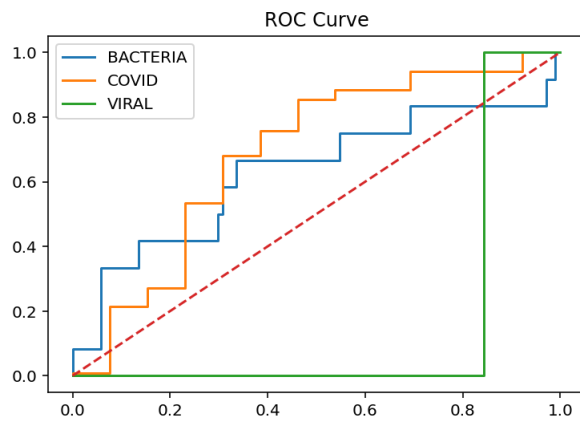


Figure 4: ROC curve for the three-class classification background-dependent model

Through the analysis of the saliency map, it was possible to identify that the model had learned to classify the images by focusing primarily on the background. Unless otherwise specified, all the updates reported below refer to a single chest X-ray scan used as an example (see Fig. 5).

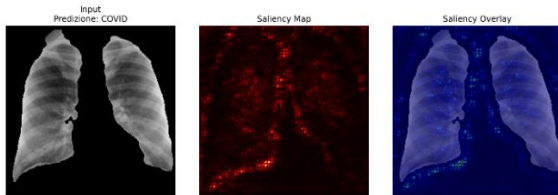


Figure 5: From the left, input, saliency map and COVID class' overlay

This image corresponds to a patient suffering from bacterial pneumonia, which was incorrectly classified as being caused by COVID-19.

Consequently, a study was conducted to determine which region of background pixels had the greatest influence on classification. It was observed that, when the mean pixel value of the lung region was used as the background value,

once the distance from the lungs exceeded 36 pixels, the probability assigned by the model changed: an incorrect classification as COVID shifted to a correct classification as BACTERIA. This allowed the region of pixels primarily responsible for the incorrect classification to be identified.

In order to understand more specifically what was influencing the classifier during training, the background was replaced using the mean pixel value of the lung region. As a result, it was observed that A remained almost unchanged, while the remaining performance metrics and the ROC curve deteriorated. It was therefore concluded that changing the background color in this manner leads to worse results. As further confirmation of the model's dependence on the background, the percentages of pixel relevance for the classification were examined for 10 samples. The first five samples were BACTERIA cases incorrectly classified as COVID, while the remaining five were samples correctly classified as COVID.

For all three classes, a predominant percentage of relevance was consistently observed in the background (see Table 3 and Fig. 6), as expected.

	BACTERIA	COVID	VIRAL
Lung core	13.98%	14.17%	17.13%
Inner border	17.23%	16.58%	19.86%
Outer border	20.70%	20.03%	23.30%
Background	48.09%	49.22%	39.71%

Table 3: Pixel relevance percentages for classification in the indicated regions, using the mean pixel value of the lungs as the background



Figure 6: Saliency map of a scan, using the mean pixel value of the lungs as the background

Through an ablation test, it was observed that the classifier exploited the background to increase the probability assigned to the COVID class, regardless of whether COVID was the correct class.

In the end, the best results were obtained by randomizing all background pixels during the training phase. This drastically reduced the classifier's dependence on the background, as demonstrated by the saliency map (see Fig. 7). Furthermore, the ROC curve became acceptable again for the COVID and BACTERIA classes, while the VIRAL class was excluded from the analysis due to the presence of only a single sample in the test set (see Fig. 8).

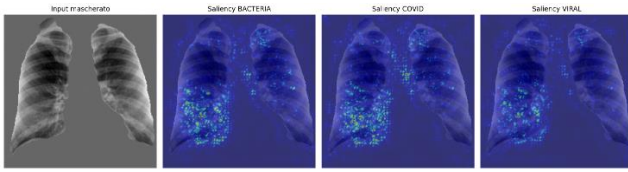


Figure 7: Saliency map of a scan, after having trained the model with randomized background

	Precision	Recall	F1 score	support
BACTERIA	0.2	0.08	0.12	12
COVID	0.89	0.96	0.93	103
VIRAL	0.00	0.00	0.00	1

Table 4: Performance metrics, after having trained the model with randomized background

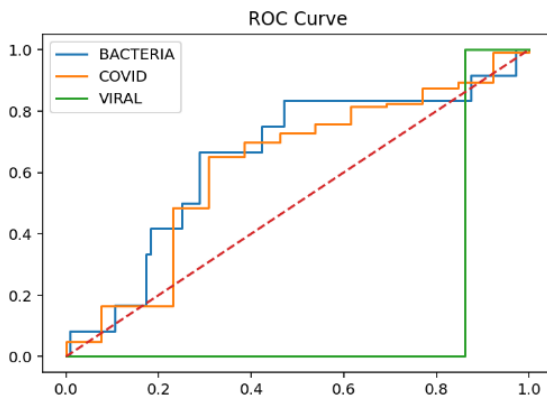


Figure 8: ROC curve, after having trained the model with randomized background

In conclusion, the three-class classifier proved unable to correctly classify the test samples into their respective actual categories while maintaining consistent performance across the classes. The results shown in Fig. 8 and Table 4 do not show any improvement over those obtained with the previous versions (see Table 1 and Fig. 4). However, by monitoring the saliency maps, it was confirmed that, with the background randomized, the model performed classification by assigning greater importance to pixels associated with the lungs.

Given the presence of only a single viral case in the test set, it was decided to proceed with a binary classification task.

5.4. Binary Classification: Preliminary Results

In an initial training and validation phase, the results shown in Fig. 9, Fig. 10 and Fig. 11 were obtained:

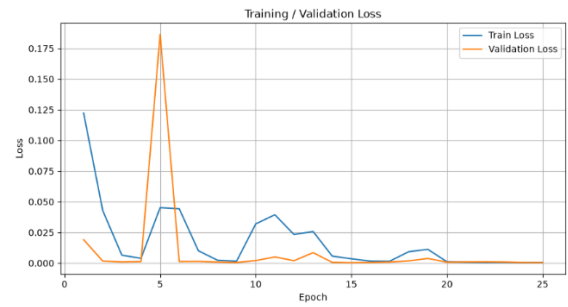


Figure 9: Training loss and validation loss trend using the initial dataset

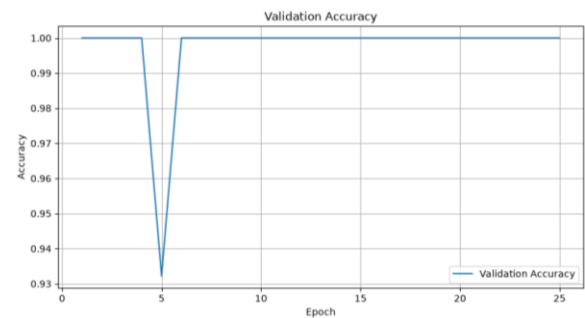


Figure 10: Validation accuracy trend using the initial dataset.

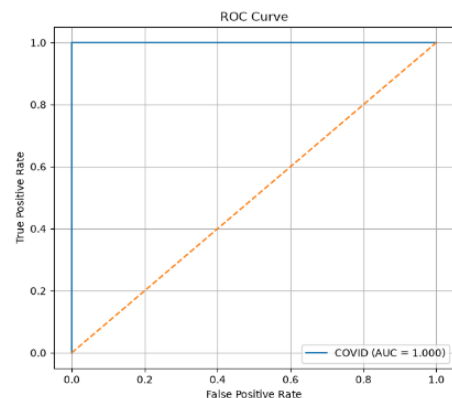


Figure 11: ROC curve using the initial dataset

This high performance motivated a further investigation aimed at assessing the possible presence of biases exploited by the network for classification, rather than information actually related to the lung region. To this end, while keeping the classifier architecture unchanged, an additional experiment was conducted in which, instead of using the region inside the lung mask, the network was trained exclusively with the portion of the image outside the mask. This made it possible to determine whether the model was still able to distinguish between the two classes in the absence of information contained within the lungs.

The results obtained with this configuration were comparable to those of the previous experiment, showing that the model was able to achieve good performance even when using exclusively the region outside the lungs (see Appendix 7.1).

Since this behavior was unexpected, was formulated the hypothesis that the images might contain features or artifacts outside the lung region capable of providing useful information for classification. For this reason, an additional verification of the correct alignment between the images and their corresponding lung masks was performed, in order to rule out any alignment or segmentation issues. As an example, an image-mask overlay is shown below:

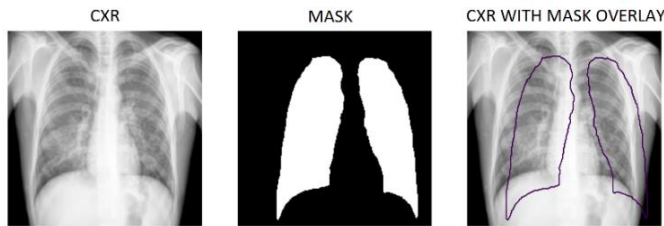


Figure 12: Visual check about mask and lungs overlapping, related to a patient affected by *Pneumocystis pneumonia*

As can be observed in Fig. 12, the shape of the lung region identified by the segmentation is correct, but the mask appears to be shifted respect to the original radiographic image. For this reason, an additional step was introduced during the preprocessing phase to correct the misalignment between the image and the mask. The correction was performed by determining the required displacement based on the distribution of pixel gradients, with the aim of achieving greater overlap between the mask and the corresponding lung region.

5.4.1. Realignment of lung masks

For comparison, an overlay of the image and one of the shifted masks is shown below. The image is the same as the one presented in Fig. 12, and it was chosen in order to demonstrate how the realignment algorithm works:



Figure 13: Lung and mask overlapping before and after algorithm application

After applying the mask realignment procedure, a visual inspection of the images was performed to assess the correctness of the identified shifts. This inspection showed that the algorithm was not always able to determine the correct alignment: in some cases, the mask was shifted to positions that did not match the outline of the lungs. This may be due to the fact that the implemented algorithm estimates

a score based on the Sobel operator, which allows regions characterized by changes in pixel intensity to be identified. Consequently, the translation may sometimes be directed towards regions other than the lung contours, since other areas of the image may also exhibit intensity variations.

Following this inspection, 143 masks in the training set were identified as requiring realignment, of which 142 belonged to the NO COVID class and 1 to the COVID class. In the test set, 27 masks were realigned, of which 26 belonged to the NO COVID class and 1 to the COVID class. The strong prevalence of NO COVID images among those subjected to correction therefore represented a potential source of bias, as the presence of a shifted mask could be correlated with the class label and provide the network with indirect information useful for classification, independently of the content of the lung region.

Once the mask positions had been corrected, the model training and validation procedures were repeated.

5.5. Binary Classification: Results with a Black Background

This section presents the training and validation results after applying the lung mask, without modifying the background.

5.5.1. Training Results

The following graphs and results were obtained.

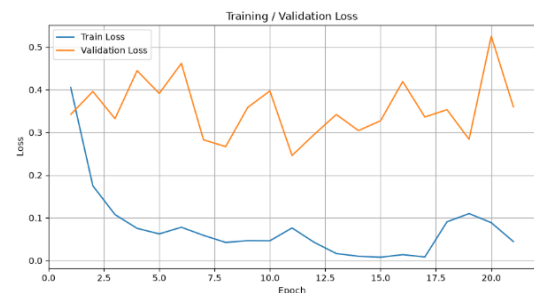


Figure 14: Training and validation loss of the binary classification model trained with a black background

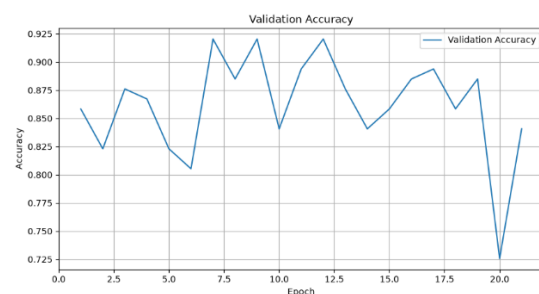


Figure 15: Validation accuracy of the binary classification model trained with a black background

During training, a progressive reduction in the training loss can be observed, reaching a minimum value of 0.0079. The validation loss, on the other hand, shows a more variable trend and reaches a minimum value of 0.2460. In particular, from epoch 11 onward, the further improvement in performance on the training set is not accompanied by a similar improvement on the validation set. This behavior is consistent with a progressive adaptation of the model to the training data and may therefore represent a possible sign of overfitting. The validation accuracy also shows some variability; however, considering that the validation set consists of a relatively limited number of images, approximately 110 samples, even a small change in the number of incorrectly classified samples can lead to noticeable variations in accuracy.

5.5.2. Validation Results

Evaluation on the test set makes it possible to determine whether the observed divergence between training and validation performance actually results in reduced generalization performance. The classifier achieves an overall accuracy of 85.4%, with the following confusion matrix:

		PREDICT	
		NO COVID	COVID
REAL	NO COVID	23	3
	COVID	12	65

Table 5: Confusion matrix of the binary classification model trained with a black background

The analysis of the classes and the different performance metrics highlights differences in classification performance. For the COVID class, the precision is 0.96, while the recall is 0.84. For the NO COVID class, the recall reaches 0.88, while the precision is 0.66. This latter value indicates that a non-negligible proportion of the images classified as NO COVID actually belongs to the COVID class, as indicated by the presence of false positives for the NO COVID class. The F1-scores are 0.90 for COVID and 0.75 for NO COVID.

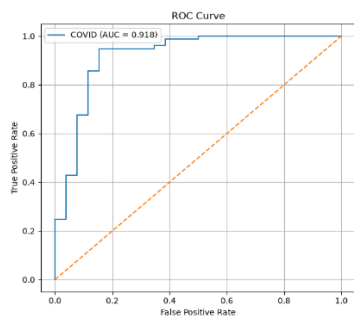


Figure 16: ROC curve of the binary classification model trained with a black background

The ROC curve also showed good discriminative ability of the classifier, with a ROC-AUC value of 0.918. This value indicated that the model was able to effectively distinguish between the two classes. The AUC value, together with the metrics obtained on the test set, therefore indicated that the possible overfitting observed during training did not prevent the model from maintaining a reasonable ability to generalize to data not used during training.

As previously mentioned, during the testing phase, an analysis based on saliency maps was performed. The saliency distribution was then analyzed across different image regions defined using the lung segmentation masks. For each image in the test set, the percentage of the total saliency associated with each region was calculated. The mean (μ) and standard deviation (σ) were then calculated for each region and for the two classes considered. This analysis made it possible to investigate how the model's saliency was distributed across the image regions and to compare its behavior between the two classes.

Region	$\mu_{NO\ COVID} \pm \sigma_{NO\ COVID}$	$\mu_{COVID} \pm \sigma_{COVID}$
Lung core	$(16.08 \pm 5.28) \%$	$(16.36 \pm 5.35) \%$
Inner border	$(10.82 \pm 1.61) \%$	$(10.82 \pm 1.57) \%$
Outer border	$(21.99 \pm 1.78) \%$	$(21.86 \pm 1.82) \%$
Background	$(51.12 \pm 6.23) \%$	$(50.96 \pm 6.28) \%$
Outside lung	$(73.11 \pm 6.26) \%$	$(72.82 \pm 6.28) \%$

Table 6: Regional analysis of saliency maps

The saliency map analysis highlights a very similar distribution of the model's saliency for the two classes considered. For both classes, approximately 73% of the total saliency is located outside the lung region, while approximately 27% falls within the lungs. Moreover, the greatest variability across images is observed for the lung core and the background, whereas the border regions show lower dispersion.

The prevalence of saliency in regions outside the lungs may indicate a possible dependence of the model on extra-pulmonary information. As an example, a saliency map obtained from one of the images is shown below:



Figure 17: Example of a saliency map obtained for a patient with bacterial pneumonia (*Streptococcus*), using the model trained with a black background

As a consequence of this dependence, the experiment was repeated using a different background.

5.6. Binary Classification: Results with Randomized Background

In the case of the randomized background, the modification was applied exclusively to the training set, while a uniform background obtained from the mean pixel values within the lung regions was maintained for the validation and test sets. This choice was made to keep the validation and testing phases fixed and controlled, preventing the randomness introduced by the randomized background from affecting the evaluation of the model's performance and ensuring the reproducibility of the results.

5.6.1. Training Results

The training results obtained are shown in Fig. 18 and Fig. 19:



Figure 18: Training e validation loss of the binary classification model trained with a randomized background

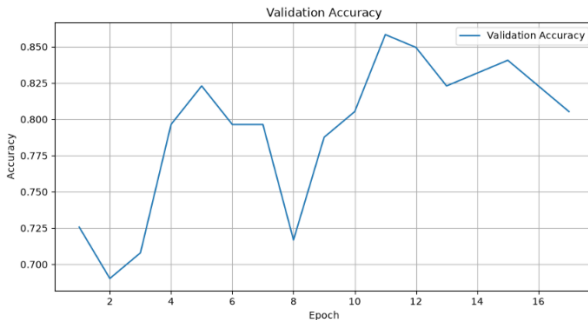


Figure 19: Validation accuracy of the binary classification model trained with a randomized background

The experiment shows that the classifier is still able to learn a discriminative representation, reaching a maximum validation accuracy of 85.8%. However, the marked reduction in training loss, accompanied by a more oscillatory trend in validation loss, highlights a discrepancy between training performance and generalization. The model selected through early stopping corresponds to epoch 7, characterized by the minimum validation loss, equal to 0.3884.

5.6.2. Validation Results

For the validation phase, the following results were obtained:

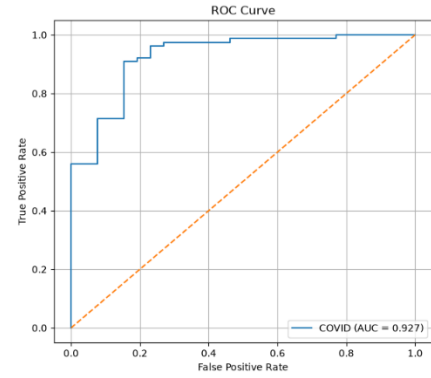


Figure 20: ROC curve of the binary classification model trained with a randomized background

The experiment conducted with a randomized background during the training phase shows good discriminative ability of the model. An accuracy of 89.3% was achieved on the test set, while the corresponding confusion matrix is reported below:

		PREDICT	
		NO COVID	COVID
REAL	NO COVID	20	6
	COVID	5	72

Table 6: Confusion matrix of the binary classification model trained with a randomized background

Furthermore, the performance differs between the two classes: the recall is 93.5% for COVID and 76.9% for NO COVID. This difference is also reflected in the other classification metrics: for the COVID class, a precision of 0.92 and an F1-score of 0.93 are obtained, whereas for the NO COVID class, these values are 0.80 and 0.78, respectively. Moreover, the ROC curve shows a ROC-AUC value of 0.927, indicating a good ability of the model to distinguish between the two classes across different decision thresholds. However, it is important to consider that the test set is not balanced, with 77 COVID images and 26 NO COVID images.

To evaluate the spatial behavior of the network, the distribution of the saliency maps was also analyzed. Compared with the previous experiment, in which the background was not randomized during training, a greater concentration of the model's saliency is observed within the lung region.

Region	$\mu_{NO\ COVID} \pm \sigma_{NO\ COVID}$	$\mu_{COVID} \pm \sigma_{COVID}$
Lung core	$(24.68 \pm 5.92) \%$	$(23.75 \pm 5.65) \%$
Inner border	$(18.43 \pm 2.45) \%$	$(18.26 \pm 2.62) \%$
Outer border	$(22.92 \pm 2.09) \%$	$(23.24 \pm 2.21) \%$
Background	$(33.97 \pm 5.53) \%$	$(34.75 \pm 5.52) \%$
Outside lung	$(56.89 \pm 6.47) \%$	$(57.99 \pm 6.40) \%$

Table 7: Regional analysis of saliency maps

In particular, the mean percentage of saliency associated with the lung core increases from approximately 16% to 24.68% for NO COVID and 23.75% for COVID, while the saliency associated with the inner border increases from approximately 11% to 18.43% and 18.26%, respectively. Conversely, the saliency associated with the background decreases and, consequently, the overall component located outside the lungs (*outside lung*) decreases from approximately 73% to 56.89% for NO COVID and 57.99% for COVID. The saliency associated with the outer border, on the other hand, remains substantially unchanged.

The saliency distribution is also very similar between the two classes, with no relevant differences across the different regions considered.

Overall, the comparison with the previous experiment indicates that randomizing the background during training leads the network to exhibit a greater proportion of saliency within the lung regions and a smaller proportion in the background.

The saliency map corresponding to the same image shown previously, but with the randomized background, is reported below:

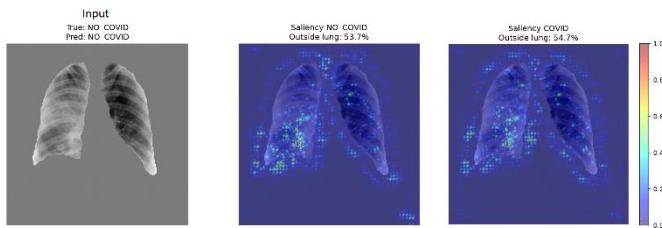


Figure 21: Example of a saliency map obtained for a patient with bacterial pneumonia (*Streptococcus*), using the model trained with a randomized background.

6. Conclusion

Initially, a three-class classification task (COVID-19, bacterial pneumonia, and viral pneumonia) was investigated. However, this approach proved ineffective due to overfitting, likely related to the imbalanced distribution of data among the three classes. For this reason, the analysis subsequently focused on binary classification. In this configuration, the DenseNet121 network, combined with the U-Net/VAE segmentation algorithm, achieved high classification performance. The analysis of saliency maps proved useful for identifying background-related biases, while randomizing the background during training substantially reduced the model's reliance on extra-pulmonary features, resulting in a greater concentration of saliency within the lung regions.

7. APPENDIX

7.1. Results Obtained Using Images with the Lung Region Subtracted

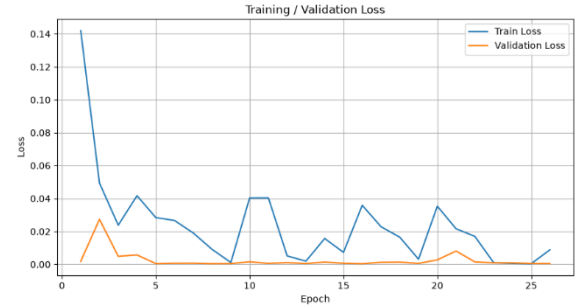


Figure 22: Training and validation loss of the binary classification model using images with the lung region removed

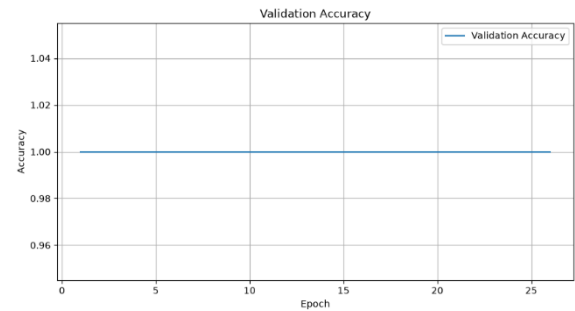


Figure 23: Validation accuracy of the binary classification model using images with the lung region removed

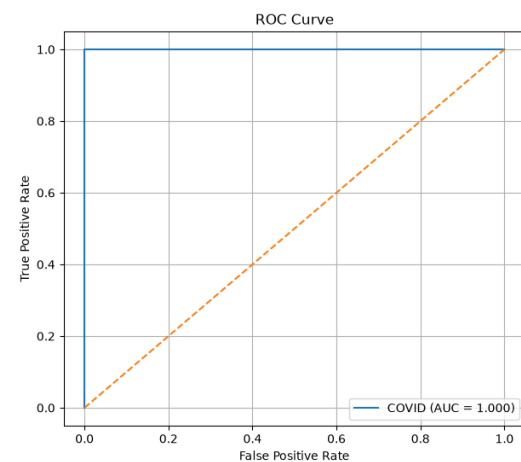


Figure 24: ROC curve of the binary classification model using images with the lung region removed